IMPACT OF COOLING CHANNEL SHAPE ON THE PERFORMANCE OF HOLLOW HAIRPIN WINDINGS

Multiphysics analysis and multi-objective channel shape optimization for direct oil-cooled hairpin windings

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Motivation

- Traction motors: higher speed and frequency
- Hairpin windings = industry standard
- AC losses → severe local hot spots
- Water jacket: long thermal path

Direct oil cooling

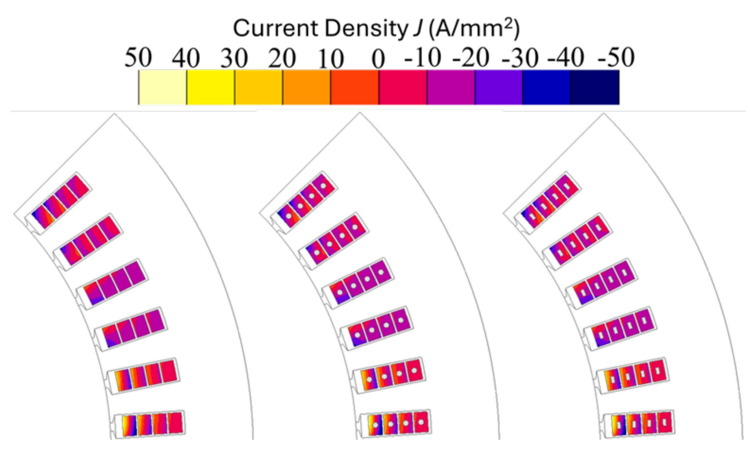
- Oil flows inside each conductor
- Heat extracted at the source
- Hot spots suppressed
- +277 % continuous current

Reference machine

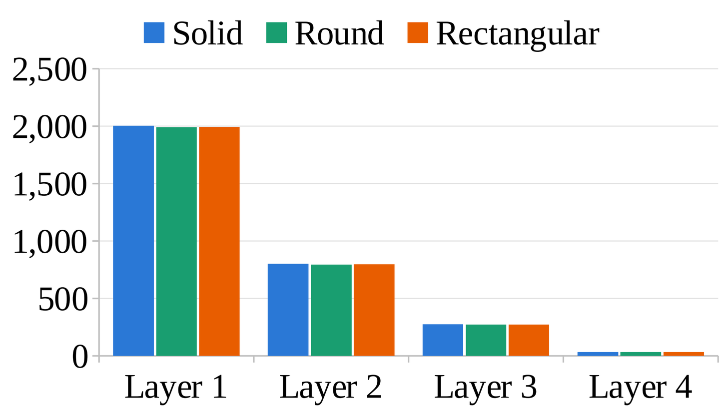
Power / speed	300 kW / 2000 rpm
Slots × layers	48 × 4
Conductor	6 × 4 mm
Current / frequency	350 A, 266.7 Hz
Coolant	Oil, 65 °C, 10 L/min
Flow regime	Laminar

Electromagnetic analysis

- 2D FEA: 350 A rms, 266.7 Hz
- Current crowds at the periphery
- Channel sits in the “dead zone”



(a) solid (b) round (c) rectangular

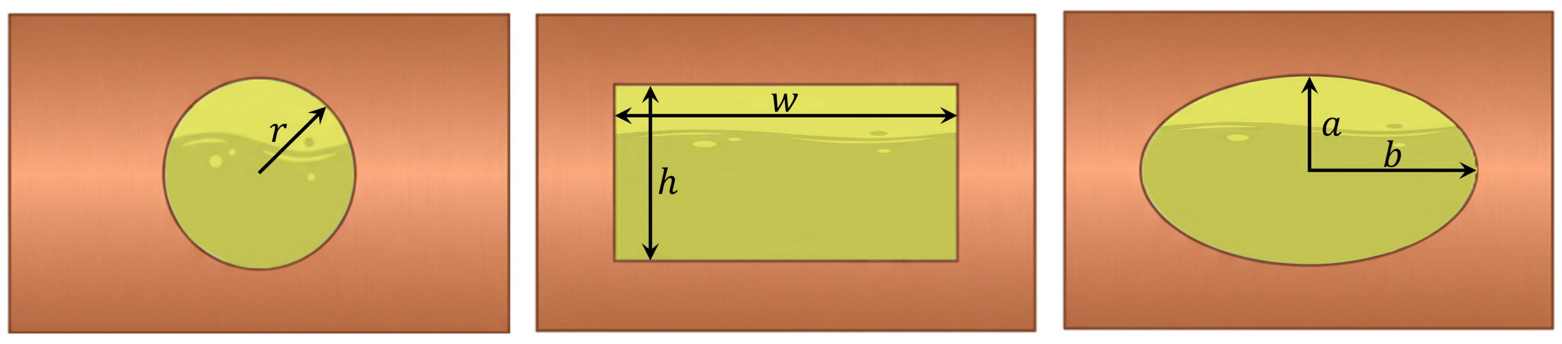


AC loss per layer (W)

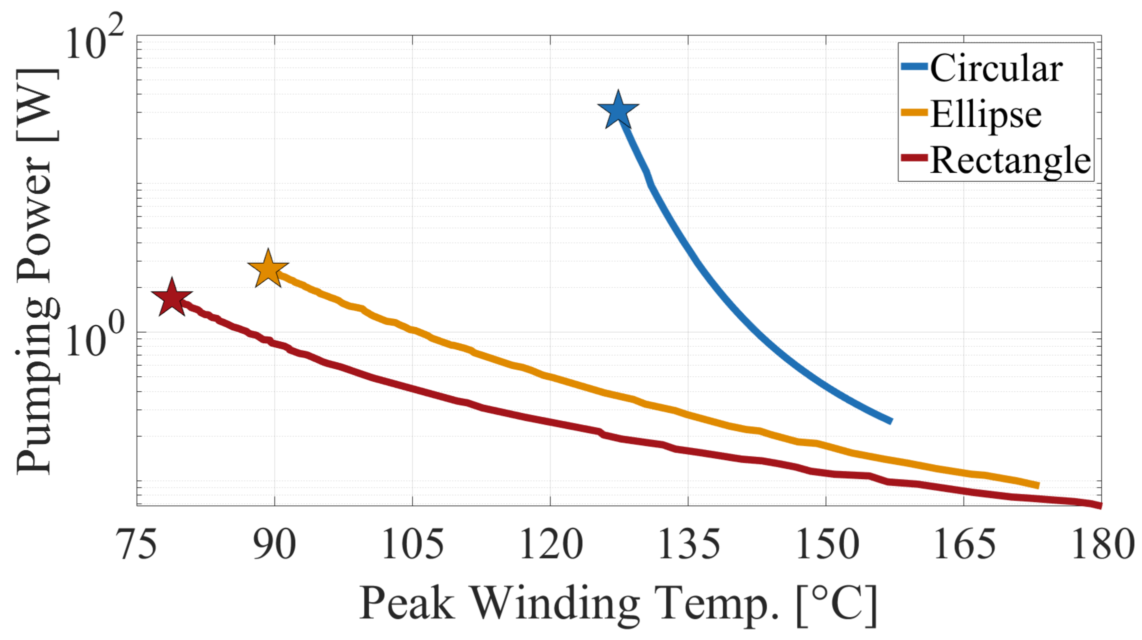
Can the channel be shaped freely for cooling?

Channel-shape optimization

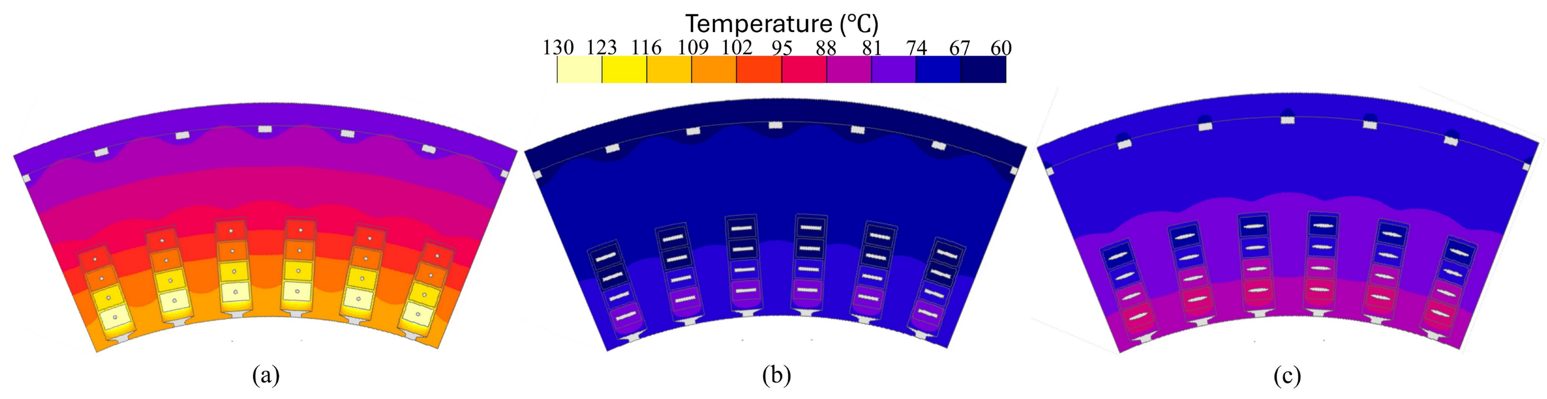
- Bigger channel: more surface, less copper
- Smaller channel: steep pressure drop
- Temperature vs. pumping power compete
- 128 FE samples → surrogate → GA
- Wall and opening ≥ 0.75 mm



(a) (b) (c)



Shape	Optimal size	Peak T	Pumping
Circular	Ø 0.75 mm	127.4 °C	34.8 W
Elliptical	4.5 × 0.75 mm	89.3 °C	2.98 W
Rectangular	**4.5 × 0.75 mm**	**78.8 °C**	**1.92 W**



Optimal designs: (a) circular (b) rectangular (c) elliptical

Conclusions

- Channel centre is EM “dead”: < 0.25 %
- Shape it purely for cooling
- Rectangle > ellipse ≫ circle
- Round → rectangular: −10 °C peak
- Optimum: 78.8 °C at 1.92 W
- Next: manufacturing + motorette test

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